

IV Calculation Formulas

A compact setup-and-unit checklist for pump, gravity, and completion-time problems.

Pump - $\text{mL/hr} = \text{total mL} / \text{hours}$. - Convert minutes to a fraction of an hour before calculating. - $\text{Completion hours} = \text{remaining mL} / \text{mL/hr}$.	Gravity - $\text{gtt/min} = (\text{mL} \times \text{drop factor}) / \text{total minutes}$. - Drop factor comes from the tubing package. - Round to whole drops unless instructed otherwise.
Safety check - Write units at every step. - Estimate before calculating. - Trace the line, verify concentration, pump library, and independent-check requirements.	

Reviewed 2026-07-13 - Source: AHRQ Patient Safety Network - <https://psnet.ahrq.gov/primer/medication-administration-errors>
Study aid only. Verify current references, instructor guidance, scope, orders, and facility policy.